

Supplementary Material for: Surrogate Inverse-Variance Weighting for Intervention Harvesting on Heavy-Tailed Click Logs

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Scope of this document. This document supplements the CIKM 2026 short paper of the same title. Theorem numbers referenced here correspond to the main paper as follows: Proposition 1 below is the *conditional delta-method variance* (Prop. “delta”, `prop:delta`); Theorem 1 is *count-only IVW on the surrogate* (Thm. “IVW”, `thm:ivw`); Theorem 2 is the *Cauchy–Schwarz surrogate guarantee* (Thm. “CS”, `thm:cs`); and Theorem 3 is the *exact asymmetric oracle* (Thm. “asym”, `thm:asym`). We restate each statement and give the complete proof, then collect the additional sensitivity analyses that the short paper defers here. Full runnable code, unit tests, and the data-processing pipelines that reproduce every number below are available at <https://github.com/alemagnani/surrogate-ivw-intervention-harvesting>.

Notation. We match the short paper exactly. q is a query, d a document/ad, k and k' result positions, and i indexes the (q, d) cells in the interventional set (the pairs that appeared at both k and k'). Under the Position-Based Model (PBM), $P(\text{click} \mid q, d, k) = p_k r_{q,d}$ with $p_k \in (0, 1]$ the position- k examination probability and $r_{q,d} = r_i \in [0, 1]$ the per-cell relevance; the true per-cell click probability is $\theta_k^i = p_k r_i$. We observe N_k^i impressions and C_k^i clicks for cell i at position k , with empirical rate $\hat{\theta}_k^i = C_k^i / N_k^i$. The target is the position-bias ratio $R_{k,k'} = p_k / p_{k'}$. Per-cell weights are $\omega_i \geq 0$ and depend only on the counts, $\omega_i = \omega(N_k^i, N_{k'}^i)$. Write

$$\hat{R}_{k,k'} = \frac{A}{B} = \frac{\sum_i \omega_i \hat{\theta}_k^i}{\sum_i \omega_i \hat{\theta}_{k'}^i}, \quad V_r(\omega) = \frac{\sum_i \omega_i^2 / N_r^i}{(\sum_i \omega_i)^2},$$

$\alpha = (1 - p_k) / p_k$, $\beta = (1 - p_{k'}) / p_{k'}$, and the harmonic weight $\omega_i^h = N_k^i N_{k'}^i / (N_k^i + N_{k'}^i)$. Throughout, the sum $\sum_i$ runs over the n cells of the interventional set, all of which have $N_k^i, N_{k'}^i \geq 1$.

1 Complete Proofs

1.1 Conditional delta-method variance

Proposition 1 (Conditional delta-method variance; Prop. delta). *Given the vector of impression counts $\mathbf{N} = \{N_k^i, N_{k'}^i\}$, the delta method gives*

$$\text{Var}(\hat{R}_{k,k'} \mid \mathbf{N}) \approx R_{k,k'}^2 [\alpha V_k(\omega) + \beta V_{k'}(\omega)],$$

with $V_r(\omega) = \sum_i \omega_i^2 / N_r^i / (\sum_i \omega_i)^2$, $\alpha = (1 - p_k) / p_k$, and $\beta = (1 - p_{k'}) / p_{k'}$.

Proof. Condition on the counts $\mathbf{N}$ throughout; all expectations and variances below are conditional on $\mathbf{N}$. Under PBM, $C_r^i \mid \mathbf{N} \sim \text{Binomial}(N_r^i, \theta_r^i)$ with $\theta_r^i = p_r r_i$, independently across cells i and

across the two positions $r \in \{k, k'\}$ (clicks at distinct impressions are independent Bernoulli draws). Hence

$$\mathbb{E}[\hat{\theta}_r^i | \mathbf{N}] = \theta_r^i, \quad \text{Var}(\hat{\theta}_r^i | \mathbf{N}) = \frac{\theta_r^i(1 - \theta_r^i)}{N_r^i}.$$

Write $A = \sum_i \omega_i \hat{\theta}_k^i$ and $B = \sum_i \omega_i \hat{\theta}_{k'}^i$, with conditional means $\bar{A} = \sum_i \omega_i \theta_k^i$ and $\bar{B} = \sum_i \omega_i \theta_{k'}^i$. Because A and B are sums of independent terms and the weights are $\mathbf{N}$ -measurable,

$$\text{Var}(A | \mathbf{N}) = \sum_i \omega_i^2 \frac{\theta_k^i(1 - \theta_k^i)}{N_k^i}, \quad (1)$$

$$\text{Var}(B | \mathbf{N}) = \sum_i \omega_i^2 \frac{\theta_{k'}^i(1 - \theta_{k'}^i)}{N_{k'}^i}. \quad (2)$$

Clicks at position k and at position k' are conditionally independent given the counts, so $\text{Cov}(A, B | \mathbf{N}) = 0$. The first-order delta expansion of $g(A, B) = A/B$ about $(\bar{A}, \bar{B})$ gives $\partial g / \partial A = 1/\bar{B}$ and $\partial g / \partial B = -\bar{A}/\bar{B}^2$, whence

$$\text{Var}\left(\frac{A}{B} \mid \mathbf{N}\right) \approx \frac{\text{Var}(A | \mathbf{N})}{\bar{B}^2} + \frac{\bar{A}^2}{\bar{B}^4} \text{Var}(B | \mathbf{N}) = \left(\frac{\bar{A}}{\bar{B}}\right)^2 \left[\frac{\text{Var}(A | \mathbf{N})}{\bar{A}^2} + \frac{\text{Var}(B | \mathbf{N})}{\bar{B}^2} \right]. \quad (*)$$

Under PBM the ratio of conditional means is exactly the estimand: $\bar{A}/\bar{B} = (p_k \sum_i \omega_i r_i) / (p_{k'} \sum_i \omega_i r_i) = p_k/p_{k'} = R_{k,k'}$, so the prefactor is $R_{k,k'}^2$.

It remains to reduce the bracket to $\alpha V_k + \beta V_{k'}$. Adopt the relevance-agnostic surrogate that drops the r_i -dependent factor from the cell-level variance, replacing $\theta_r^i(1 - \theta_r^i)$ by the common position-level factor $\theta_r^i/N_r^i \cdot (\dots)$; concretely, using $\theta_r^i = p_r r_i$ and factoring r_i out of both $\text{Var}(A | \mathbf{N})/\bar{A}^2$ and the mean, the leading term of $\text{Var}(A | \mathbf{N})/\bar{A}^2$ is

$$\frac{\sum_i \omega_i^2 \theta_k^i(1 - \theta_k^i)/N_k^i}{(\sum_i \omega_i \theta_k^i)^2} \xrightarrow{\text{surrogate}} \frac{1 - p_k}{p_k} \cdot \frac{\sum_i \omega_i^2/N_k^i}{(\sum_i \omega_i)^2} = \alpha V_k(\omega),$$

obtained by writing $\theta_k^i(1 - \theta_k^i) = p_k r_i(1 - p_k r_i)$, approximating $1 - p_k r_i \approx 1 - p_k$ (equivalently, treating r_i as a common factor across cells, which is exact when r_i is constant) and cancelling the common $p_k r_i$ between numerator and the squared mean. The identical reduction at position k' gives $\beta V_{k'}(\omega)$. Substituting into $(*)$ yields $\text{Var}(\hat{R}_{k,k'} | \mathbf{N}) \approx R_{k,k'}^2 [\alpha V_k(\omega) + \beta V_{k'}(\omega)]$. The exact (non-surrogate) bracket, retaining the r_i -dependent factors, is Eq. (3) of the short paper ($\tilde{V}$); the delta step $(*)$ and the conditional independence used there are identical. $\square$

Remark 1. The factors $\alpha = (1 - p_k)/p_k$ and $\beta = (1 - p_{k'})/p_{k'}$ are the per-position binomial overdispersion terms. Because p_k is larger for higher (more examined) positions, α is smaller near the top of the ranking. When $p_k \approx p_{k'}$ (smooth position-bias decay across adjacent positions) we have $\alpha \approx \beta$, and the conditional variance collapses to $\alpha [V_k(\omega) + V_{k'}(\omega)]$: the symmetric surrogate of Theorems 1 and 2.

1.2 Count-only IVW optimum on the symmetric surrogate

Theorem 1 (Count-only IVW on the surrogate; Thm. IVW). *Among all non-negative weights $\omega_i = \omega(N_k^i, N_{k'}^i)$, harmonic weighting minimises the count-only delta-method variance surrogate $V_k(\omega) + V_{k'}(\omega)$, and*

$$\min_{\omega \geq 0} [V_k(\omega) + V_{k'}(\omega)] = \left(\sum_i \omega_i^h \right)^{-1}.$$

Proof. Combine the two positions into a single quadratic form. With $a_i := 1/N_k^i + 1/N_{k'}^i > 0$,

$$V_k(\omega) + V_{k'}(\omega) = \frac{\sum_i \omega_i^2 (1/N_k^i + 1/N_{k'}^i)}{(\sum_i \omega_i)^2} = \frac{\sum_i a_i \omega_i^2}{(\sum_i \omega_i)^2}.$$

Set $u_i = \omega_i \sqrt{a_i}$ and $v_i = 1/\sqrt{a_i}$, so that $\sum_i \omega_i = \sum_i u_i v_i$ and $\sum_i a_i \omega_i^2 = \sum_i u_i^2$. The Cauchy–Schwarz inequality $(\sum_i u_i v_i)^2 \leq (\sum_i u_i^2)(\sum_i v_i^2)$ gives

$$V_k(\omega) + V_{k'}(\omega) = \frac{\sum_i u_i^2}{(\sum_i u_i v_i)^2} \geq \frac{1}{\sum_i v_i^2} = \frac{1}{\sum_i 1/a_i},$$

with equality iff $u_i \propto v_i$, i.e. $\omega_i \propto 1/a_i$. Now

$$\frac{1}{a_i} = \frac{1}{1/N_k^i + 1/N_{k'}^i} = \frac{N_k^i N_{k'}^i}{N_k^i + N_{k'}^i} = \omega_i^h,$$

so the minimiser is $\omega_i \propto \omega_i^h$, harmonic weighting. The estimator $\hat{R}_{k,k'} = A/B$ is invariant to a common positive scaling of the weights, so any $\omega_i \propto \omega_i^h$ produces the same estimator; harmonic is therefore *the* count-only optimum. Finally the minimum value equals $1/\sum_i (1/a_i) = 1/\sum_i \omega_i^h$, which also matches the collapsed left-hand side used in Theorem 2. $\square$

1.3 Cauchy–Schwarz surrogate guarantee (unconditional in counts)

Theorem 2 (Surrogate guarantee, unconditional in counts; Thm. CS). *For any impression counts $\{N_k^i, N_{k'}^i\}_{i=1}^n$,*

$$V_k(\omega^h) + V_{k'}(\omega^h) \leq V_k(\mathbf{1}) + V_{k'}(\mathbf{1}),$$

i.e. harmonic weighting never increases the symmetric surrogate relative to uniform ($\omega_i \equiv 1$). Equality holds iff $N_k^i/N_{k'}^i$ is constant across all cells i .

Proof. The key algebraic identity is that the harmonic weight is precisely the reciprocal of the per-cell coefficient $a_i = 1/N_k^i + 1/N_{k'}^i$:

$$(\omega_i^h)^2 \left(\frac{1}{N_k^i} + \frac{1}{N_{k'}^i} \right) = (\omega_i^h)^2 \cdot \frac{N_k^i + N_{k'}^i}{N_k^i N_{k'}^i} = (\omega_i^h)^2 \cdot \frac{1}{\omega_i^h} = \omega_i^h. \quad (\text{collapse})$$

Substituting $\omega = \omega^h$ into the left-hand side and writing $W = \sum_i \omega_i^h$,

$$V_k(\omega^h) + V_{k'}(\omega^h) = \frac{\sum_i (\omega_i^h)^2 (1/N_k^i + 1/N_{k'}^i)}{W^2} = \frac{\sum_i \omega_i^h}{W^2} = \frac{W}{W^2} = \frac{1}{W}.$$

For the uniform weights $\omega_i \equiv 1$ (so $\sum_i \omega_i = n$),

$$V_k(\mathbf{1}) + V_{k'}(\mathbf{1}) = \frac{1}{n^2} \sum_i \left(\frac{1}{N_k^i} + \frac{1}{N_{k'}^i} \right) = \frac{1}{n^2} \sum_i \frac{1}{\omega_i^h},$$

using $1/N_k^i + 1/N_{k'}^i = 1/\omega_i^h$ again. Apply Cauchy–Schwarz to the vectors $(\sqrt{\omega_i^h})_i$ and $(1/\sqrt{\omega_i^h})_i$:

$$n^2 = \left(\sum_i \sqrt{\omega_i^h} \cdot \frac{1}{\sqrt{\omega_i^h}} \right)^2 \leq \left(\sum_i \omega_i^h \right) \left(\sum_i \frac{1}{\omega_i^h} \right) = W \sum_i \frac{1}{\omega_i^h}.$$

Dividing by n^2W gives $\frac{1}{W} \leq \frac{1}{n^2} \sum_i \frac{1}{\omega_i^h}$, i.e. $V_k(\omega^h) + V_{k'}(\omega^h) \leq V_k(\mathbf{1}) + V_{k'}(\mathbf{1})$. Cauchy–Schwarz holds with equality iff the two vectors are proportional, i.e. ω_i^h is constant across i ; since $\omega_i^h = N_k^i/(1 + N_k^i/N_{k'}^i)$ and the estimator is scale-invariant, this reduces (up to the overall scale that cancels) to $N_k^i/N_{k'}^i$ being constant across cells, the “no count-heterogeneity” case in which harmonic coincides with uniform. $\square$

Remark 2 (From the surrogate to $\text{Var}(\hat{R} \mid \mathbf{N})$). *Theorem 2 is unconditional in the counts but conditional on $V_k + V_{k'}$ being the objective. Via Proposition 1 it implies a reduction in $\text{Var}(\hat{R} \mid \mathbf{N})$ whenever $\alpha \approx \beta$. For the asymmetric objective $\alpha V_k + \beta V_{k'}$ the same substitution gives only an empirical bound in general (the unconditional analogue can fail; see §2.4), but on the full KDD log the asymmetric reduction (79.0%) essentially matches the symmetric one (79.3%).*

1.4 Exact asymmetric oracle

Theorem 3 (Exact asymmetric oracle; Thm. asym). *Fix any counts $\{N_k^i, N_{k'}^i\}$ and any $\alpha, \beta > 0$, and set $s_i = \alpha/N_k^i + \beta/N_{k'}^i > 0$. Then for all non-negative weights ω ,*

$$\alpha V_k(\omega) + \beta V_{k'}(\omega) \geq \left(\sum_i s_i^{-1} \right)^{-1},$$

with equality iff $\omega_i \propto \omega_i^ := 1/s_i = N_k^i N_{k'}^i / (\alpha N_{k'}^i + \beta N_k^i)$. The symmetric harmonic weight is the special case $\alpha = \beta$.*

Proof. Collect the two positions into a single quadratic form with the asymmetric coefficient s_i :

$$\alpha V_k(\omega) + \beta V_{k'}(\omega) = \frac{\sum_i \omega_i^2 (\alpha/N_k^i + \beta/N_{k'}^i)}{(\sum_i \omega_i)^2} = \frac{\sum_i s_i \omega_i^2}{(\sum_i \omega_i)^2}.$$

Apply Cauchy–Schwarz to the vectors $(\sqrt{s_i} \omega_i)_i$ and $(1/\sqrt{s_i})_i$:

$$\left(\sum_i \omega_i \right)^2 = \left(\sum_i \sqrt{s_i} \omega_i \cdot \frac{1}{\sqrt{s_i}} \right)^2 \leq \left(\sum_i s_i \omega_i^2 \right) \left(\sum_i \frac{1}{s_i} \right).$$

Rearranging,

$$\frac{\sum_i s_i \omega_i^2}{(\sum_i \omega_i)^2} \geq \frac{1}{\sum_i 1/s_i} = \left(\sum_i s_i^{-1} \right)^{-1},$$

with equality iff $\sqrt{s_i} \omega_i \propto 1/\sqrt{s_i}$, i.e. $\omega_i \propto 1/s_i$. Computing the reciprocal,

$$\frac{1}{s_i} = \frac{1}{\alpha/N_k^i + \beta/N_{k'}^i} = \frac{N_k^i N_{k'}^i}{\alpha N_{k'}^i + \beta N_k^i} = \omega_i^*,$$

which is the claimed minimiser. When $\alpha = \beta$, $s_i = \alpha(1/N_k^i + 1/N_{k'}^i)$, so $1/s_i \propto N_k^i N_{k'}^i / (N_k^i + N_{k'}^i) = \omega_i^h$: the oracle reduces to harmonic weighting. $\square$

Corollary 1 (Oracle unconditionally beats uniform). *For all counts and all $\alpha, \beta > 0$, $\alpha V_k(\omega^*) + \beta V_{k'}(\omega^*) \leq \alpha V_k(\mathbf{1}) + \beta V_{k'}(\mathbf{1})$.*

Proof. The uniform value is $\alpha V_k(\mathbf{1}) + \beta V_{k'}(\mathbf{1}) = (1/n^2) \sum_i s_i$. Cauchy–Schwarz gives $n^2 \leq (\sum_i s_i)(\sum_i 1/s_i)$, hence $(1/n^2) \sum_i s_i \geq (\sum_i 1/s_i)^{-1} = \alpha V_k(\omega^*) + \beta V_{k'}(\omega^*)$ by Theorem 3. $\square$

Remark 3 (Why the oracle is not deployable, and harmonic is). ω_i^* depends on the unknown propensities through (α, β) , so it cannot be computed from counts alone. Harmonic weighting is the parameter-free choice: exact at $\alpha = \beta$ (Theorem 3), with an unconditional symmetric guarantee for all propensity values (Theorem 2). Empirically, substituting the true simulation propensities into ω_i^* yields variance within $\pm 2\%$ of harmonic across all tested configurations, and a two-stage plug-in oracle ($\hat{\alpha}, \hat{\beta}$ from a preliminary harmonic fit) matches harmonic and the true-propensity oracle to within 0.4% (§2.5). The exact θ -aware oracle that plugs in per-cell $\hat{r}_i$ is a different object: on heavy-tailed logs it collapses (ARE 0.002) because $\hat{r}_i$ is unusable on the singleton mass, which is why the count-only surrogate is the robust choice.

2 Additional Sensitivity Analyses

All numbers in this section are computed by the public repository. Where a specific figure is a deterministic functional of the observed KDD counts (variance surrogates, ARE, ESS, capping) it is reproducible exactly; where it depends on Monte-Carlo resampling (bootstrap CIs, trial-averaged MSE) we report the seed-controlled values used in the paper.

2.1 Weight-capping robustness (ARE vs percentile cap)

The harmonic weight concentrates mass on a small set of high-count cells: on the full KDD (1, 2) interventional set (4.61M pairs) the harmonic effective sample size is only $\text{ESS} = 1,849$ ($\text{ESS}(\omega) = (\sum_i \omega_i)^2 / \sum_i \omega_i^2$). This concentration is exactly what produces the surrogate- $\tilde{V}$ reduction, and the natural robustness probe is to cap the largest weights. Capping ω_i^h at a percentile threshold and recomputing the plug-in ARE against uniform gives Table 1.

Table 1: Weight-capping robustness on full KDD (1, 2). Capping ω_i^h at the given percentile lifts ESS toward the uniform value but retains a clear ARE advantage over uniform at every level. The 99th-percentile cap shifts the estimand ($\hat{R} : 1.71 \rightarrow 1.61$), consistent with the precision/estimand trade-off in the main paper.

Cap level	threshold / # capped	ESS	ARE(harmonic, uniform)
None (harmonic)	–	1,849	4.53
99.99th pct	2,079 / 462 pairs	5.2×10^4	4.12
99th pct	–	7.6×10^5	2.83
90th pct	–	–	1.86

Harmonic beats uniform robustly across all capping levels; the advantage degrades gracefully as the cap forces the weights toward uniform. This directly addresses the concern that the estimate “rests on $\sim 1,800$ cells”: even after redistributing weight away from the top 10% of cells, harmonic retains ARE 1.86.

2.2 Per-query impression-distribution diagnostic

The main paper states that per-cell counts on the full KDD log are extremely heavy-tailed and points here for the underlying per-query/per-pair impression-distribution diagnostic. Concretely, on the full KDD Cup 2012 Track 2 release (235.6M impressions, 63.8M unique $(q, \text{ad}, \text{pos})$ cells; (1, 2) interventional set of 4.61M (q, ad) pairs, 132.3M impressions):

- **Head concentration.** The pair-size distribution has median $N_1 + N_2 = 4$ and maximum 2.47M; 60.7% of pairs have $N_1 + N_2 \leq 5$, while 0.1% of pairs carry 42% of all impressions. The maximum per-pair imbalance N_1/N_2 reaches 66,330:1.
- **Imbalance shape.** Only $\sim 65\%$ of (q, ad) items have $N_1/N_2 \in [0.5, 2.0]$ and $\sim 3\text{--}5\%$ exceed ratio 10 —the heavy-tailed regime the anchor-item synthetic model targets.
- **Contrast with deduplicated public logs.** The released Baidu-ULTR aggregation has 94.1% of (q, d, pos) cells at $N = 1$, 98.7% at $N \leq 3$, and 78.4% of adjacent-pair ratios within $[0.5, 2.0]$ (only 0.8% above 10). We attribute this to per-cell deduplication in the public release, which erases exactly the cross-document count heterogeneity that makes weighting matter; it is a property of the released dataset, not of production logs.

The diagnostic makes the “head queries hold most impressions” statement quantitative: because a small head of (q, ad) pairs holds most of the impression mass, a query-cluster bootstrap (§2.6) reflects the heavy-tailedness of the *query* distribution rather than the estimator’s conditional efficiency. The full per-pair histograms are produced by the diagnostic script in the repository.

2.3 Paired bootstrap CIs: harmonic vs. min

The short paper reports that harmonic and min are empirically tied on KDD (ARE 4.53 vs. 4.41). To quantify whether the $\sim 5\text{--}7\%$ synthetic margin of harmonic over min is statistically distinguishable, we computed paired bootstrap 95% CIs on per-trial squared-error margins with seed-aligned RNG (so per-trial outcomes are paired across schemes; split 80/20, 500 trials per cell, 2000 bootstrap resamples). Results are in Table 2.

Table 2: Paired bootstrap 95% CIs on per-trial MSE margins (split 80/20, 500 trials, 2000 resamples). ★ marks CIs excluding zero. Margins in units of 10^{-4} ; bracketed CI endpoints in the same units.

Comparison	$r = 1$	$r = 10$	$r = 50/100$
★ harmonic – original	−7.1 [−9.9, −4.9]	−6.7 [−8.0, −5.6]	−7.1 [−8.4, −5.9]
harmonic – min	−0.35 [−0.62, −0.08] ★	−0.18 [−0.36, 0.00]	not significant

The harmonic–original margin’s CI *excludes zero in every tested cell*: harmonic reliably beats uniform. The harmonic–min margin’s CI *crosses zero at $r = 50$ and $r = 100$* (and is borderline at $r = 10$): at high imbalance the harmonic-vs-min gap is not statistically distinguishable at $n = 500$ trials. We therefore do not oversell the margin over min; the case for harmonic rests on the surrogate-IVW characterisation (Theorem 1) and the unconditional guarantee (Theorem 2) that min lacks, not on an empirical margin. This is consistent with the KDD bootstrap of $\hat{R}$ itself, where harmonic and min agree within $\sim 0.06\%$ on (1, 2) while uniform diverges by $\sim 9\%$ (non-overlapping CIs).

2.4 Asymmetric surrogate: counterexample and structured search

Theorem 2 bounds the *symmetric* surrogate $V_k + V_{k'}$ unconditionally. The asymmetric analogue $\alpha V_k(\omega^h) + \beta V_{k'}(\omega^h) \leq \alpha V_k(\mathbf{1}) + \beta V_{k'}(\mathbf{1})$ does *not* hold unconditionally, and we record the failure mode explicitly.

Closed-form $n = 2$ counterexample. Take $\alpha = 100$, $\beta = 0.01$ with counts $(N_k^1, N_{k'}^1) = (100, 1)$ and $(N_k^2, N_{k'}^2) = (2, 2)$. Writing $f(\omega) = \alpha V_k(\omega) + \beta V_{k'}(\omega)$, direct computation gives $f(\omega^h) = 12.876$ against $f(\mathbf{1}) = 12.754$: harmonic is $\sim 1\%$ worse than uniform. Structurally, when $\alpha \gg \beta$ a cell with $N_k \gg N_{k'}$ has small $s_i = \alpha/N_k^i + \beta/N_{k'}^i$ and small $\omega_i^h \approx N_{k'}^i$, so $(\omega_i^h)^2$ and s_i become positively correlated rather than anti-correlated —the opposite of the covariance condition needed for $f(\omega^h) \leq f(\mathbf{1})$.

Structured two-archetype search. A random count sweep over $[1, 1000]^2$ almost never produces this pattern. Replacing it with a structured search over mixtures of two archetypes (“imbalanced” $(M, 1)$ vs. “balanced” (b_0, b_0) , sweeping the mixture fraction from 0 to 1) reveals the failure at every grid point: restricting $(\alpha, \beta) \in [0.1, 10]^2$ (covering all adjacent and non-adjacent PBM pairs with $\eta = 1$, $M = 10$), harmonic loses to uniform in 100% of a 15×15 (α, β) grid on at least one structured config, worst-case ratio 7.78; on the wider range $[0.05, 100]^2$ the worst ratio reaches 15.4.

Why this does not undermine harmonic in practice. Realistic interventional sets contain heterogeneous counts (anchors plus regular docs with per-doc fluctuations), not degenerate two-archetype mixtures. On the full KDD counts, harmonic does not underperform uniform on the asymmetric surrogate: with the marginal-CTR estimates $\hat{p}_1 = 0.0501$, $\hat{p}_2 = 0.0256$ (so $\alpha = 18.97$, $\beta = 38.08$, $\alpha/\beta \approx 0.50$), harmonic reduces $\alpha V_k + \beta V_{k'}$ by a factor 4.75 versus uniform, essentially matching the symmetric reduction of 4.84 (within 2%). The KDD per-pair counts are positively correlated (items popular at position 1 are typically also popular at position 2), the opposite of the anti-correlated pattern the counterexample requires. A meaningful *conditional* asymmetric bound (a regularity condition on the count distribution under which harmonic provably dominates uniform asymmetrically) remains open.

2.5 Plug-in θ -oracle / EM comparison

Two “oracle” comparisons clarify why the count-only surrogate is the right object to optimise on heavy-tailed logs.

Two-stage plug-in of the asymmetric oracle. We implemented the adaptive scheme of Theorem 3: fit harmonic to obtain preliminary $\hat{p}$, derive $\hat{\alpha}, \hat{\beta}$, and apply $\hat{\omega}_i^* = N_k^i N_{k'}^i / (\hat{\alpha} N_{k'}^i + \hat{\beta} N_k^i)$. On the synthetic benchmark (split 80/20, 500 trials), plug-in, harmonic, and the true-propensity oracle agree to within 0.4% in MSE (Table 3); the plug-in’s marginal gains (0.3–0.4% relative) are smaller than the trial-to-trial standard error. The asymmetric oracle therefore provides essentially no benefit over parameter-free harmonic, strengthening the case for harmonic.

Table 3: Plug-in oracle vs. harmonic and the true-propensity oracle, MSE (split 80/20). Plug-in does not statistically improve over harmonic; both match the true oracle to within trial noise.

$r =$	10	50	100
original	1.67×10^{-3}	1.58×10^{-3}	1.49×10^{-3}
harmonic	7.87×10^{-4}	7.31×10^{-4}	7.24×10^{-4}
plug-in (this paper)	7.84×10^{-4}	7.30×10^{-4}	7.22×10^{-4}
true-propensity oracle	7.85×10^{-4}	7.30×10^{-4}	7.22×10^{-4}

θ -aware oracle and PBM-EM on KDD. The exact inverse-variance oracle uses per-cell relevance through $\theta_r^i(1 - \theta_r^i)$. Plugging in per-cell $\hat{r}_i$ on the full KDD (1, 2) set makes this θ -aware oracle *pathological* (ARE 0.002, SE 0.148, ESS ≈ 1.1): $\hat{r}_i$ is unusable on the $\sim 60\%$ singleton mass, so the oracle chases noise. Beta shrinkage of $\hat{\theta}_i$ (pseudo-counts $a \in \{1, \dots, 1000\}$) lifts its ARE only from 0.002 to 0.61 —still below harmonic’s 4.53. Relatedly, vanilla no-feature PBM-MLE (one r_i per cell) on KDD (1, 2) collapses to $\hat{R}_{1,2} = 1.682$, the uniform estimand, because the model is over-parameterised on singleton-dominated data; feature-based EM [1] avoids this but needs features absent from the anonymised aggregation. On clean synthetic PBM data the same MLE recovers $R = 2.0$ (SE 0.035; harmonic 0.040, uniform 0.055; 500 trials), confirming the implementation. The comparison between model-free intervention harvesting and parametric joint inference is thus methodologically ill-posed rather than merely omitted: the two paradigms solve different estimation problems.

2.6 Query-cluster (cluster) bootstrap

The click-resample (parametric Bernoulli bootstrap conditional on counts) is the empirical baseline the plug-in $\tilde{V}$ is validated against, agreeing to within 1–2%; it isolates the conditional-on-counts click-noise efficiency that Theorem 2 bounds. A *query-cluster bootstrap* resamples entire queries (clusters of (q, ad) pairs) rather than individual pairs, and therefore measures variation in the underlying *query population*. Because a small head of queries holds most impressions (§2.2), the cluster bootstrap is dominated by that heavy-tailed query distribution: it is a diagnostic of the log, not of the estimator at fixed counts, and it penalises any concentrated-weight (inverse-variance-style) estimator through the effective-sample-size mechanism familiar from importance sampling. This is the same reason naive pair-bootstrap is misaligned as an efficiency diagnostic here: on full KDD, harmonic/min show $\sim 3\times$ *larger* pair-bootstrap std than uniform on (1, 2), not because they are less precise conditional on counts (they reduce the surrogate by 79%), but because resampling the count population perturbs the high-information cells that drive any inverse-variance estimator. We report the cluster/pair bootstrap for transparency; the load-bearing real-data result is the deterministic 79% surrogate reduction on the observed counts.

References

- [1] X. Wang, M. Bendersky, D. Metzler, and M. Najork. Position bias estimation for unbiased learning to rank in personal search. In *WSDM*, 2018.